LAB 2

# GEOSPATIAL REPRESENTATION LEARNING

## Lab 2: Geodata Scavenger Hunt

PRESS → OR SPACE. →

© Marc Rußwurm
Licensed under **Creative Commons Attribution–NonCommercial 4.0 International (CC BY-NC 4.0)**.
You may share and adapt these slides for teaching, research, and other non-commercial educational purposes with attribution.
Commercial training, paid workshops, consulting seminars, or incorporation into commercial course products require prior permission.

UNIVERSITÄT BONN

# The Challenge

## Find one way of measuring our area of interest on Earth.

Each of you will acquire one different geospatial data modality
and contribute it to our shared class dataset.

# Same Place — Many Measurements

**AOI** →

- 📡 radar backscatter
- 🛰️ optical reflectance
- 🏔️ elevation
- 👥 population
- 🌊 soil moisture
- 🌡️ surface temperature
- 🌧️ precipitation
- 🌱 land cover
- 🛣️ road network
- *… and more*

The same coordinates can carry many different measurements. Each student contributes one layer.

# Your Assignment

**Every student receives:**

- One assigned data modality or source
- The same Area of Interest (AOI)
- The same required output CRS

Work on your assigned source only.

**Assignment table**

⚠ *Instructor: insert or link to the student–dataset assignment table here.*

AOI: *[ to be provided ]*
Required output CRS: *[ e.g. EPSG:XXXXX ]*
Shared Drive folder: *[ link ]*
Shared Google Slides: *[ link ]*

# What You Need to Deliver

① One geospatial data file covering the AOI

② Data kept at its **native spatial resolution** — do not resample

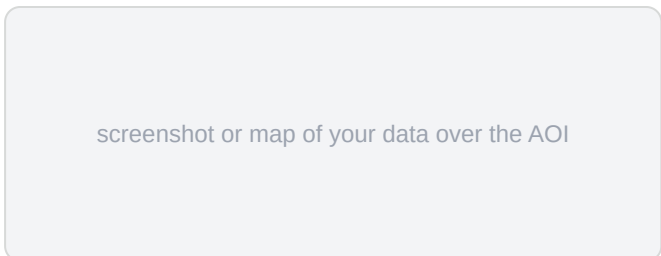
③ Data reprojected to the required output CRS

④ File uploaded to the shared Google Drive folder

⑤ One completed slide in the shared Google Slides deck

# Your Slide (Template)

**[ Dataset / product name ]**




**What does it measure?** …
**Provider / source** …
**Native spatial resolution** …
**Temporal reference** …
**Native CRS** …
**Final CRS** …
**File format** …

---

**How did you get it?** (2–3 short steps)
**One difficulty or surprise** (one sentence)

# The Workflow

**Find** → **Understand** → **Download** → **Clip** → **Reproject** → **Check** → **Upload**

**Before downloading anything, determine:**

- What does this dataset actually represent?
- What is its CRS?
- What is its spatial resolution?
- What is its temporal meaning?
- What are the nodata / missing-value conventions?

# Coordinate Systems Matter

source dataset

**native CRS**

(e.g. geographic WGS 84)

↓

coordinate transformation

required output

**class CRS**

*[ EPSG:XXXXX ]*

Reprojection changes the **coordinate representation**.
It should not change the **spatial resolution**.

After reprojection: verify that the data still overlaps the AOI.

Plot it. Do not assume.

# Native Resolution Is Intentional

**Do not make the datasets look the same yet.**

| | | |
|---|---|---|
| **elevation**<br>~30 m | **optical imagery**<br>10 m | **population**<br>100 m |
| **land cover**<br>native categorical grid | **climate**<br>kilometres | *your modality* |

The resolution mismatch is part of the problem.

# A Teaser for Lab 3

### Continuous values

- elevation
- surface temperature
- spectral reflectance

interpolation between pixels is meaningful

### Categorical values

- land-cover class
- soil type

interpolation between pixels is not meaningful

**What should the value be between two pixels?**
This depends on what the values mean.

# If You Get Stuck

1. Read the dataset / product documentation
2. Check the CRS and spatial extent of the file you downloaded
3. Inspect the downloaded file locally — open it, plot it
4. Ask a neighbour what they would check first
5. Ask the instructor

Do not spend 45 minutes fighting a broken download portal.
If a source is inaccessible or genuinely broken, flag it early.

# Timeline

| | |
|---|---|
| 00:00–00:10 | Briefing and assignments |
| 00:10–00:25 | Find your source — understand the product and its metadata |
| 00:25–01:10 | Download / access the data |
| 01:10–01:30 | Clip to AOI, reproject to class CRS, verify |
| 01:30–01:40 | Upload data file and complete your slide |
| 01:40–02:10 | **Lightning presentations** |

# Lightning Presentations

## 60–90 seconds per student

**Answer only:**

1. What does your dataset measure?
2. Where did you get it?
3. What did you have to do to make it usable?
4. What was the hardest or most surprising part?

No deep dives into the science behind the modality.

No live demos unless everything already works.

The variety of acquisition stories — APIs, portals, registration walls, 4 GB files — is itself the learning content.

# What We Built Together

one place

**AOI**

+

N independent modalities

**your datasets**

=

shared result

**multimodal geospatial dataset**

Keep your data and code. We will use them again in Lab 3.

**Next step:** coordinates → sampled values → machine-learning dataset